days. This position of the earth is called the summer solstice.

On 22nd December, the Tropic of Capricorn receives direct rays of the sun as the south pole tilts towards it. As the sun's rays fall vertically at the Tropic of Capricorn (23%° s), a larger portion of the southern hemisphere gets light. Therefore, it is summer in the southern hemisphere with longer days and shorter nights. The reverse happens in the northern hemi-sphere. This position of the earth is called the winter solstice.

Statement 2: INCORRECT—Equinox

On 21 March and 23 September, direct rays of the sun fall on the equator. At this position, neither of the poles is tilted towards the sun; so, the whole earth experiences equal days and equal nights. This is called an equinox.

Statement 3 and 4: CORRECT—Days and nights and changes in the seasons because of the rotation and revolution of the earth respectively.

33. **Statement 1: CORRECT**—The atmosphere absorbs certain frequencies and lets through some other frequencies of solar radiation. In other words, the atmosphere regulates the entry of solar radiation.

    Statement 2: CORRECT—Harmful ultraviolet radiation would find its way through, if the atmosphere (ozone in stratosphere to be specific) were absent.

    Statement 3: CORRECT—The atmosphere also takes care of extra-terrestrial objects like meteors which get burnt up while passing through the atmosphere (mesosphere to be precise) due to friction.

34. **Statement 1: INCORRECT**—Ozone is unevenly distributed. It is between 20 km and 25 km altitude that the greatest concentrations of ozone are found.

    Statement 2: CORRECT—It is formed at higher altitudes and transported downwards.

    Statement 3: INCORRECT—The ozone layer is typically thicker over the poles than over the equator for three reasons. First, there is a lack of sunlight during an arctic winter to break it down. Second, seasonal weather systems and wind patterns in the upper